

VacuPowerlift

Tireless and powerful

Structure and application

The VacuPowerlift is a modular product system that can be easily and quickly adapted to a wide variety of applications. It essentially consists of a rotary head with an integrated check valve, a telescopic lifting tube with multiple winding layers, and a suction pad — together forming the lifting unit. The control unit is mounted on the suction pad and can be connected using extensions and joints.

To grip different workpieces, a wide range of suction grippers can be attached to the lifting unit. These not only have to function using vacuum technology but can also be pneumatically operated thanks to an integrated compressed-air feedthrough. In addition, pivoting and rotating devices are available to significantly simplify handling.

If all of this is still not sufficient, customized gripping tools and special applications tailored to your needs can be developed. This enables you to optimize your production process while protecting your employees from long-term injuries through ergonomically designed workstations.

The application areas of the VacuPowerlift are virtually unlimited. It is used across many industrial and trade sectors and combines safe handling with ergonomic operation for loads of up to 300 kg. At the same time, it simplifies production processes while offering high cost efficiency and rapid amortization.

- Cartons in shipping and logistics
- Drums and containers in the chemical and pharmaceutical industries
- Panel handling in wood and sheet-metal processing
- Stones and concrete parts in the construction industry
- Loading and unloading of CNC machining centers
- Bags in agriculture and the bulk materials industry

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VacuPowerlift
Assembly overview

Vacuum generator



Vacuum pumps

For the VPL 100–VPL 140 series, dry-running vacuum pumps with 25–40 m^3/h and an integrated vacuum filter are available.



Sound-insulated pumps

The dry-running vacuum pumps are also available in a sound-insulated version, providing significant noise reduction.



Vacuum blowers

The robust and powerful side-channel blowers are available in three sizes with a suction volume of 130–280 m^3/h.



Additional silencers

Additional silencers can be mounted on the discharge side of the vacuum blowers, reducing the noise level by approximately 3–4 dB(A).



Sound insulation box

The vacuum blowers can be installed in sound insulation boxes that reduce the noise level by up to 10 dB(A). Also available as a retrofit option.



Vacuum filter for blowers

Large, high-performance vacuum filter with high flow rate to protect the vacuum blowers from contamination.

VacuPowerlift
Assembly overview

Lifting units



Rotary head

All rotary heads feature high-performance bearings as well as an integrated check valve that ensures slow lowering, for example in the event of a power failure.



Lifting tube

The telescopic lifting tubes of the tube lifters are made from multi-layer wound materials that ensure a long service life.



Suction pad

The robust and stable steel design ensures a secure connection of the various gripping units.



Pneumatic module

With the integrated pneumatic feed-through, pneumatic gripping tools and pivoting devices can also be supplied.

VacuPowerlift
Assembly overview

Control housing and grip options



Standard control housing

With support handle and control bar for intuitive, easy operation.



Radio control housing

With integrated push button for the optional wireless remote control of the vacuum generator.



Pneumatic control housing

With integrated pneumatic hand valve for the easy control of pneumatic movements.



Universal control housing

Version with multiple actuation options for a wide range of functions depending on requirements.



Rigid extension

For safe operation of large-format workpieces.



Articulated extension

For an ergonomic operating position when handling workpieces from high and low lifting heights.



Quick venting

Additional venting valve for faster aeration and release of non-porous workpieces. Easy operation via the hand lever.



Balancer

For balancing the control housing when using long extensions or heavier additional options.

VacuPowerlift
Assembly overview

Additional functions



Quick-change coupling

Enables the replacement of suction grippers for different applications during ongoing operation.



Endless rotating unit

Layer-by-layer, offset restacking or simple rotation of workpieces. Also available with a detent module for 90° locking.



Manual pivoting unit

Simple and fast tilting device for pivoting smaller workpieces up to approx. 35 kg.



Pneumatic pivoting unit

Pivoting device driven by one or two pneumatic cylinders for workpieces up to 150 kg.



Pneumatic turning unit

Pneumatic device for 180° turning. The height of the turning bracket is adjusted to the size of the workpiece.



Wall or column brackets

For floor-free installation of the vacuum generators on walls or columns.



Compensating valve

A compensating valve is available for the use of mechanical gripping tools, providing automatic regulation of the vacuum.

VacuPowerlift
Assembly overview

Gripping systems



Single suction gripper

In round, flat, or bellows design – suitable for drums, containers, and small cartons.



Single suction gripper

In a flat, rectangular design, ideal for workpieces with narrow suction surfaces.



Bale suction gripper

With two flat suction grippers in a special reinforced rubber design.



Bag suction gripper

Plastic version, available in two sizes for small and large paper and plastic bags.



Double suction gripper

With round and rectangular suction grippers – ideal for cartons.



Quadruple suction gripper

With round and rectangular suction grippers, ideal for large-area panel materials.



Canister gripper

Pneumatically equipped clamping gripper for simultaneously relocating up to 20 canisters at once.



Canister gripper

Simple hook gripper for undergripping the canister tab, equipped with a load-balancing valve for vacuum control.



Box gripper

For pneumatic clamping and pivoting of plastic crates.

VacuPowerlift
Assembly overview

Gripping systems



Drum gripper

With pneumatic clamping and pivoting device for emptying filled drums.



Pallet gripper

With mechanical undergrippers for handling pallets.



Magnetic gripper

With pneumatic actuation for handling perforated sheets.



Area gripper

Equipped with an area suction gripper and integrated ejector.



Combination gripper

Needle and magnetic gripper for insulation materials and perforated sheets. Actuation is carried out via two pneumatic valves.



Pneumatic gripper

Equipped with a clamping device for relocating hollow blocks.



Pneumatic gripper

With internal clamping jaws for relocating slit strips.



Suction gripper for long goods

With a reinforced longitudinal beam design for handling long goods up to 6,000 mm.



Sixfold suction gripper

For handling large-area and thin-walled panel materials.

VacuPowerlift
Assembly overview

Stainless steel and ATEX version

For the food, pharmaceutical, and chemical industries, the VacuPowerlift is also available in a stainless steel and ATEX version. All components are made of V2A or—where higher requirements apply—V4A stainless steel and are fully welded. For ATEX requirements, special blowers and electrically conductive device versions are available, ensuring safe operation in ATEX zones 21 and 22.



Stainless steel lifting unit

The rotary head, suction pad, and control housing are manufactured entirely in stainless steel. For ATEX requirements, the lifting tube features a special antistatic coating and includes an open, internal copper wire insert.



ATEX lifting unit

For ATEX requirements, the lifting tube is equipped with a special antistatic coating and also contains an open, internal copper wire insert. All components are electrically conductively connected, ensuring the required equipotential bonding.

VacuPowerlift
Assembly overview



Suction grippers

A wide variety of suction grippers is available, all of which are made of stainless steel.



ATEX vacuum blowers

Specially designed vacuum blowers in ATEX version for zones 21 and 22.



Ejektor

Pneumatic vacuum generator for EX zones 1 and 2.



ATEX vacuum filter

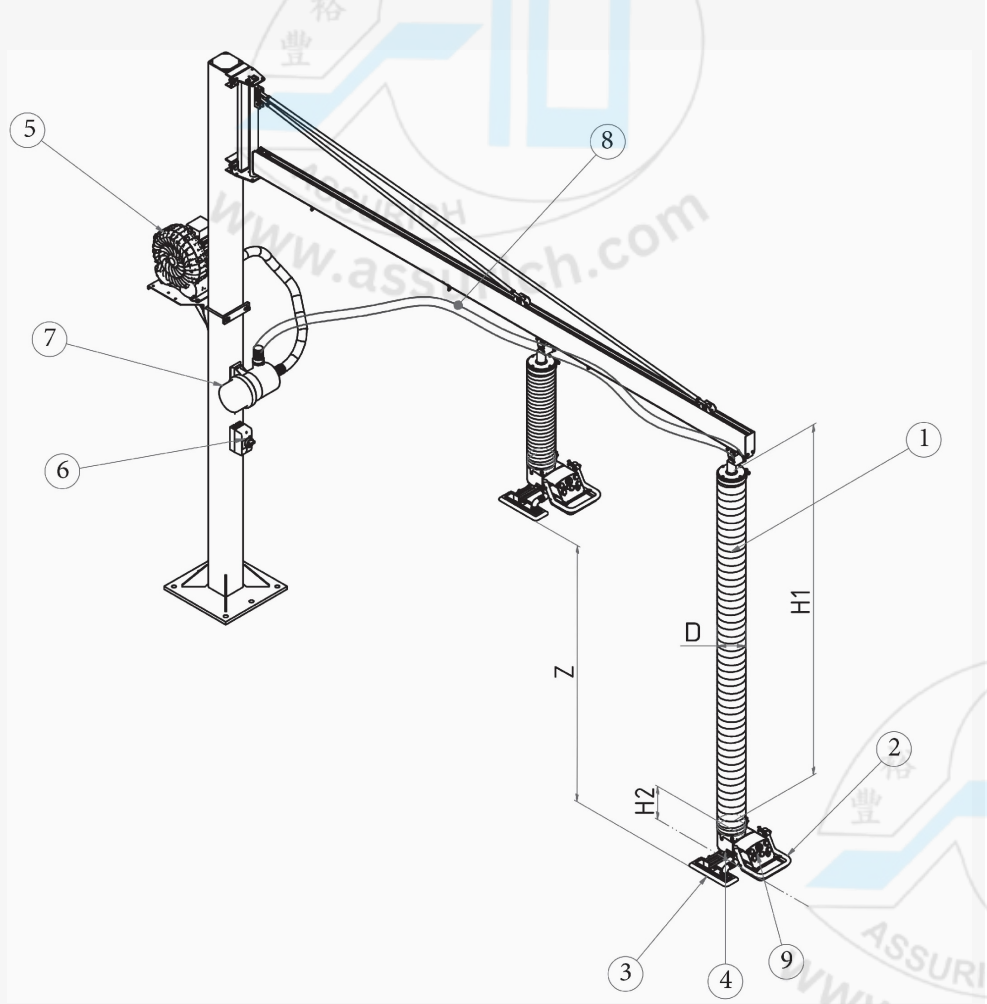
Special stainless steel filter with an internal filter cartridge and electrically conductive connection to the housing.



Accessories

The most common accessories, such as quick-change devices or control handle extensions, are available in stainless steel versions.

VacuPowerlift
Technical data



VacuPowerlift
Technical data

1 Lifting unit consisting of rotary head, lifting tube, and suction pad
A safety valve in the rotary head allows suction-held loads to descend in a controlled and slow manner in the event of a power failure.

2 Control unit with operating handle and control bar for the functions 'Lower', 'Lift', and 'Release'
With intuitive operation and a noise-optimized design.

3 Suction gripper or gripping unit with custom adaptations
With a safety factor of 2.5, they provide the necessary hold for the safe handling of loads.

4 Accessories

5 Robust and high-performance vacuum generators and blowers
Ensure the generation of the required vacuum and enable fast handling processes.

6 Motor protection switch or radio remote control
Protect the vacuum generators from overload and offer the option of remote operation via the control unit.

7 Large-dimensioned vacuum filters with replaceable cartridges
Protect the vacuum generators from contamination and damage.

8 Supply hose
Connects the vacuum generators to the lifting unit. Lengths of up to 50 meters and more are possible.

9 Adjustment screw for working height
Allows suction-held loads to be positioned at a predefined height to perform additional work steps.

Type	Load capacity non-porous/ porous (kg)	Workpiece dimensions min (mm)	Workpiece dimensions max (mm)	Vacuum generator	Lifting speed (m/min)	Overall height H1 (mm)	Control unit H2 (mm)	Rotary head H3 (mm)	Dia-meter D (mm)	Lift Z (mm)	Dead weight (kg)
VPL100	30/25	200 x 200	2.000 x 1.000	T4.40	25-55	2.715/4.215	120	95	100	1.700/2.500	6-10
VPL120	40/35	200 x 200	2.000 x 1.000	T4.40	25-55	2.715/4.215	120	95	120	1.700/2.500	6-10
VPL140	50/40	500 x 500	2.000 x 1.000	SKV300	30-60	2.715/4.215	120	95	140	1.700/2.500	8-12
VPL160	65/50	500 x 500	3.000 x 1.500	SKV300	30-60	2.730/4.230	120	110	160	1.700/2.500	8-12
VPL180	85/65	750 x 750	3.000 x 1.500	SKV400	30-60	2.730/4.230	120	110	180	1.700/2.500	12-16
VPL200	110/90	750 x 750	4.000 x 2.000	SKV400	30-60	2.730/4.230	120	110	200	1.700/2.500	12-16
VPL230	140/120	750 x 750	4.000 x 2.000	SKV400	30-60	2.730/4.230	120	110	230	1.700/2.500	14-20
VPL250	200/160	750 x 750	4.000 x 2.000	SKV400	30-60	2.730/4.230	120	110	250	1.700/2.500	14-20
VPL300	300/250	750 x 750	4.000 x 2.000	SKV500	25-50	2.730/4.230	120	110	300	1.700/2.500	18-24

*depending on the suction grippers and accessories used